

ferred to as third droop impedance. The trough close to 1MHz frequency range is governed by the bulk capacitor and other motherboard capacitors' effective capacitance.

The second crest point in the plot is dictated by the motherboard parasitic inductance and the CPU-motherboard interconnect parasitic inductance. This point is referred to as second droop impedance. The

trough following the second droop is governed by LSCs and the DSCs (die side caps).

The steep rise in the impedance leading to the third crest is dictated by package parasitic inductance and LSC/DSC parasitic inductance and is referred to as first droop impedance. The falling impedance in high frequency region > 100MHz is governed by metal-insulated-metal

(MIM) capacitances inside the silicon die.

It becomes extremely important to generate and understand the impedance profile of the system under test to run any kind of optimisation on the same. For example, addition or removal of bulk capacitor on motherboard will not have any impact on second droop impedance as it's beyond the range of

its frequency response. Package capacitors and package layout need to be optimised for reducing the same. Such design level understanding can only be achieved through system frequency domain analysis.

Transient analysis.

Transient analysis is the ultimate test for system reliability. It helps in comparing the 'final' PDN performance with device level specifications defined in the datasheets. Also, the results generated can be correlated with actual validation results to estimate the overall simulation process accuracy. It involves generating a time-domain passive circuit model of the PDN under analysis.

With circuit model generated, proper voltage source (at VR location) and sink loads (at load devices) can be connected at respective nodes of the model to generate an overall circuit file. This circuit file can be simulated and analysed for results using any SPICE based circuit simulator, such as HSPICE, LTSPICE, etc. The key results generated are the time domain voltage profiles at the input of system level

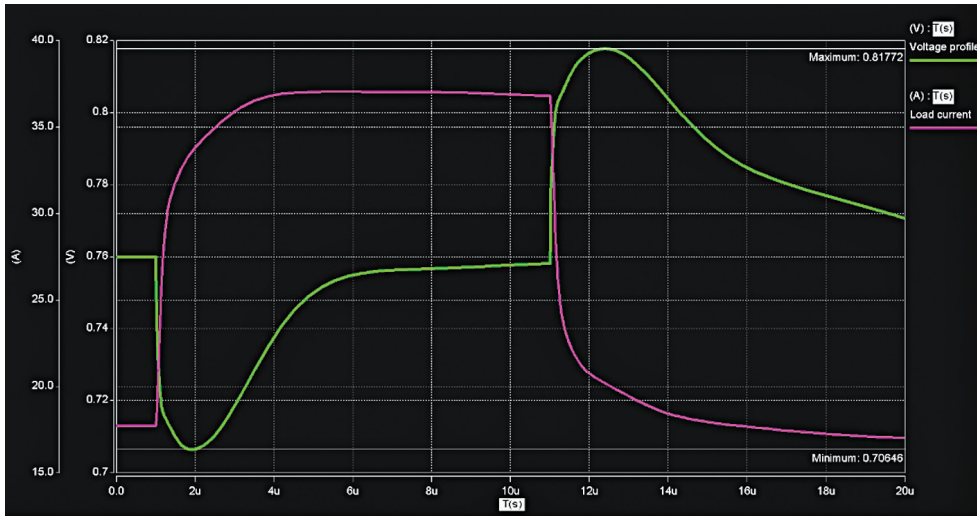


Fig. 7: Time domain voltage profile across system device under test with load profile

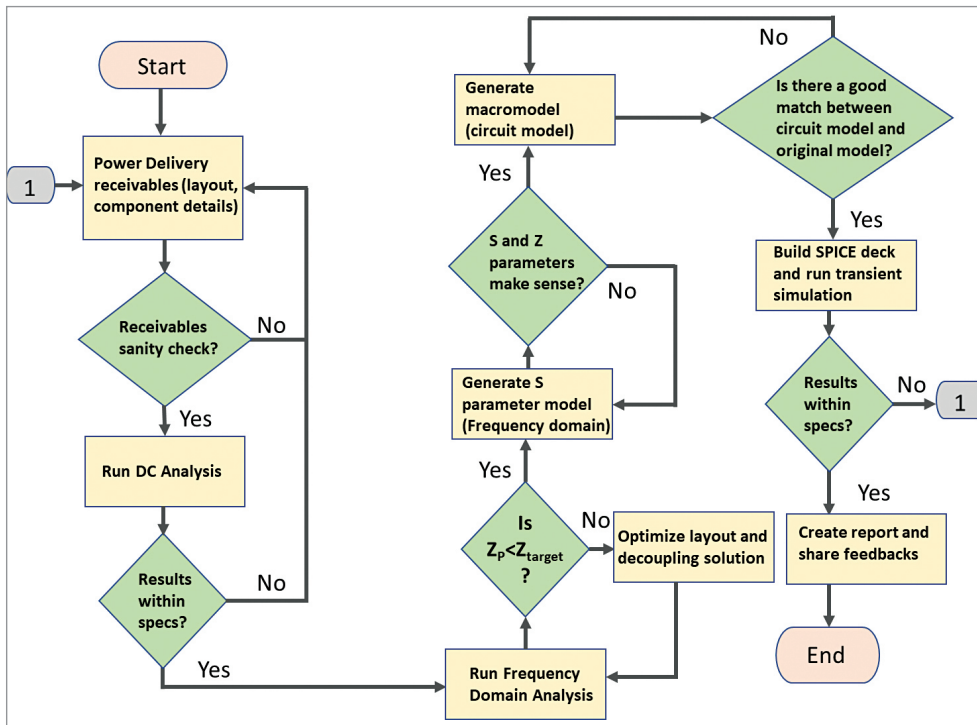


Fig. 8: Power integrity analysis workflow